# Introduction to Software Engineering

# Software Testing

The student team is required to complete the **Software Testing** documentation for the assigned course project, following the attached template.

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# Table of Contents

**Objectives** ..... **1**
**1 Member Contribution Assessment** ..... **2**
**2 Test plan** ..... **3**
**3 Test cases** ..... **4**
3.1 List of test cases ..... 4
3.2 Test case specifications ..... 6
3.2.1 Test case 1 ..... 6
3.2.2 Test case 2 ..... 6
3.2.3 Test case 3 ..... 7
3.2.4 Test case 4 ..... 7
3.2.5 Test case 5 ..... 8
3.2.6 Test case 6 ..... 8
3.2.7 Test case 7 ..... 9
3.2.8 Test case 8 ..... 9
3.2.9 Test case 9 ..... 10
3.2.10 Test case 10 ..... 10

# Software Testing

## Objectives

This document focus on the following topics:

- ✔ Completing the Software Testing document with the following sections:
  - Test Plan
  - Test Cases
- ✔ Understanding the Software Testing document.

# 1 Member Contribution Assessment

| ID | Name | Role | Contribution (%) | Signature |
|---|---|---|---|---|
| 23127061 | Trương Lý Khải | **Project Manager**<br>Business Analyst<br>Tester | 100% | |
| 23127207 | Đặng Đăng Khoa | Designer<br>Backend Developer | 100% | |
| 23127271 | Võ Ngọc Bích Trâm | Designer<br>Frontend Developer<br>Tester | 100% | |
| 23127404 | Lê Tuấn Lộc | Fullstack Developer & Engineer | 100% | |
| 23127486 | Phan Quốc Thịnh | Designer<br>Frontend Developer | 100% | |

# 2 Test plan

Our main method of verification is through **black box testing and dynamic verification**. For every use case listed in the previous documentation, a list of test cases will be created and refined. After the project is finished and can be run,, one of our two testers (who will examine the website's layout first before testing to understand the general concept) will handle each test case in the order listed within this document.

Each test case of a use case has specific **required inputs, order of steps and expected outputs**. Once the order of steps is done, if the corresponding output matches or parallels the expected output, then the test case is marked as passed, otherwise it is marked as failed. Repeat until the list of test cases are exhausted.

Certain use cases (like login/sign in) have much more test cases than others. These use cases are known to have a wide variety of situations that can happen when applied, as well as having more importance to the operation of the software (for example, a use case like "logging in" is required to access the rest of the features within the software).

Our main method of validation is through **alpha testing with a limited number of testers**. Since we already have used our team's testers for verification, validation will be handled through a selective group of alpha testers that we can trust.

These testers will help us confirm whether the software meets the description stated previously in the first documentation of the project.

# 3 Test cases

## 3.1 List of test cases

| Seq | Test case | Target | Description |
|---|---|---|---|
| 1 | Normal Login | User/Admin | Default Login Option |
| 2 | Empty Login | User/Admin | Login with missing Inputs |
| 3 | Invalid Email Login | User/Admin | Login with an invalid email address |
| 4 | Incorrect Password Login | User/Admin | Login with an incorrect password of the specified email |
| 5 | Non-existent Login | User/Admin | Login with an account that does not exist |
| 6 | Password Recovery | User/Admin | Option to recover password through email |
| 7 | Normal Sign Up | User | Default Sign up Option |
| 8 | Empty Sign Up | User | Sign up with missing Inputs |
| 9 | Invalid Email Sign Up | User | Sign up with an invalid email address |
| 10 | User Search | User | Search for the name of another User |
| 11 | Liking Posts | User | Like another User's post |
| 12 | Commenting on Posts | User | Comment under any post |
| 13 | Browsing Pokédex | User | Browse the information of all available Pokémons on the Pokédex |
| 14 | Pokédex Search Engine | User | Search Pokémons through the Pokédex |
| 15 | Pokédex Filter Engine | User | Filter selective traits of Pokémons through the Pokédex |

| | | | |
|---|---|---|---|
| 16 | Viewing Profile | User/Admin | View the profile page of the account |
| 17 | Editing Profile | User/Admin | Edit the profile page of the account |
| 18 | Following Users | User | Follow another User to see their profile page and their posts more often |
| 19 | Following Own's Account | User | Trying to follow one's own account (should result in a failure) |
| 20 | Unfollowing Users | User | Unfollow another User that was followed before |
| 21 | Reply to Comments | User | Reply to other comments in a section under a post |
| 22 | Image Compatibility | User | Profile pages and posts can have images on them |
| 23 | Map Access | User | Access the map webpage from the main webpage |
| 24 | Map Functions | User | Zoom in/out and adjust position on the map |
| 25 | Pokémon Map Tracking | User | Track the locations/pins of a Pokémon on the specified map |
| 26 | Pokémon Map Search | User | Search the Pokémon to track on the specified map |
| 27 | Pin Modification | User | Mark, unmark or edit a pin on the map |
| 28 | Warn Posts | Admin | Warn a post of a User that violates a guideline |
| 29 | Delete Posts | Admin | Delete a post of a User that majorly violates a guideline or violates one multiple times. |

| | | | |
|---|---|---|---|
| 30 | View User Lists | Admin | View the list of users currently using the website |

## 3.2 Test case specifications

### *3.2.1 Test case 1*

| *Test case* | *T001* |
|---|---|
| *Related Use case* | *Login* |
| *Context* | *Main webpage, Login Form* |
| *Input Data* | *Email: letuanloc1412@gmail.com*<br>*Password: Lok@1412* |
| *Expected Output* | *The website is updated to the main page*<br>*Login is successful* |
| *Test steps* | *1. Load the website*<br>*2. Press the "Sign in" button on the top right corner*<br>*3. Fill in the corresponding input data*<br>*4. Press the "Sign in" confirmation button* |
| *Actual Output* | *Matches expected output* |
| *Result* | *Passed* |

### *3.2.2 Test case 2*

| *Test case* | *T007* |
|---|---|
| *Related Use case* | *Register Accounts* |
| *Context* | *Main webpage, Sign in Form* |
| *Input Data* | *Email: testing1331@gmail.com*<br>*Username: test*<br>*Password: testing@1331*<br>*Confirm Password: testing@1331*<br>*OTP is a 6-digit code sent to the corresponding email* |
| *Expected Output* | *The website is updated to the sign in page.*<br>*Register successful.* |

| | |
|---|---|
| *Test steps* | *1. Load the website*<br>*2. Press the "Sign in" button on the top right corner*<br>*3. Press the "Sign up" link on the bottom*<br>*4. Fill in the corresponding input data*<br>*5. Press the "Sign up" confirmation button*<br>*6. Fill in the OTP*<br>*7. Press "Confirm"* |
| *Actual Output* | *Matches expected output* |
| *Result* | *Passed* |

### *3.2.3 Test case 3*

| *Test case* | *T010* |
|---|---|
| *Related Use case* | *Find Users* |
| *Context* | *Main webpage, after test T001* |
| *Input Data* | *Search Bar: iuem* |
| *Expected Output* | *The user "Iuem1412" shows up under the search bar* |
| *Test steps* | *1. Select the Search Bar*<br>*2. Input the corresponding input data* |
| *Actual Output* | *Matches expected output* |
| *Result* | *Passed* |

### *3.2.4 Test case 4*

| *Test case* | *T011* |
|---|---|
| *Related Use case* | *Like Posts* |
| *Context* | *Main webpage, after test T001* |
| *Input Data* | *Post: The first Post that displays status as not liked*<br>*Input: The like button (icon of a star)* |
| *Expected Output* | *The star icon glows up and the like number is incremented* |
| *Test steps* | *1. Scroll the posts list until the corresponding post is found.* |

| | |
|---|---|
| | *2. Like the post in the corresponding method.* |
| *Actual Output* | *Matches expected output* |
| *Result* | *Passed* |

### *3.2.5 Test case 5*

| *Test case* | *T012* |
|---|---|
| *Related Use case* | *Comment on Posts* |
| *Context* | *Main webpage, after test T011* |
| *Input Data* | *Post: The same post as test T011*<br>*Comment Text: "testing 123"* |
| *Expected Output* | *The comment is added to the start of the comment list* |
| *Test steps* | *1. Select the Comment icon on the post*<br>*2. Type the corresponding input data into the Comment Prompt*<br>*3. Press "Send"* |
| *Actual Output* | *Matches expected output* |
| *Result* | *Passed* |

### *3.2.6 Test case 6*

| *Test case* | *T014* |
|---|---|
| *Related Use case* | *Read Pokédex* |
| *Context* | *Pokédex webpage, after test T001* |
| *Input Data* | *Pokédex Search Bar: "Charmander"* |
| *Expected Output* | *The corresponding Pokémon is shown under the prompt* |
| *Test steps* | *1. Select the Pokédex*<br>*2. Type the corresponding input data into the Comment Prompt* |
| *Actual Output* | *Matches expected output* |
| *Result* | *Passed* |

### 3.2.7 Test case 7

| *Test case* | *T017* |
|---|---|
| *Related Use case* | *Customize Profile* |
| *Context* | *Main webpage, after test T001* |
| *Input Data* | *Gender: "Female"*<br>*Bio: "New Bio"* |
| *Expected Output* | *The corresponding profile information is updated to the user.* |
| *Test steps* | *1. Select the avatar on the top right corner*<br>*2. Select "View Profile"*<br>*3. Select "Edit Profile" button*<br>*4. Input the corresponding data*<br>*5. Press "Confirm"* |
| *Actual Output* | *Matches expected output* |
| *Result* | *Passed* |

### 3.2.8 Test case 8

| *Test case* | *T018* |
|---|---|
| *Related Use case* | *Follow Users* |
| *Context* | *Main webpage, after test T001* |
| *Input Data* | *Post: The first post of user "dangkhoa"*<br>*Follow: Following* |
| *Expected Output* | *The follow status of the chosen user is updated and stored* |
| *Test steps* | *1. Scroll the posts list until the corresponding post is found.*<br>*2. Press the "Follow" button* |
| *Actual Output* | *Matches expected output* |
| *Result* | *Passed* |

### 3.2.9 Test case 9

| *Test case* | *T023* |
|---|---|
| *Related Use case* | *Change Systems* |
| *Context* | *Main webpage, after test T001* |
| *Input Data* | *-* |
| *Expected Output* | *The map is shown to the user with the default map selected* |
| *Test steps* | *1. Select the Pokémap option on the top right*<br>*2. Wait until the map is loaded* |
| *Actual Output* | *Matches expected output* |
| *Result* | *Passed* |

### 3.2.10 Test case 10

| *Test case* | *T025* |
|---|---|
| *Related Use case* | *Track Pokémons* |
| *Context* | *Map webpage, after test T023* |
| *Input Data* | *Tracked Pokémon: #12 Butterfree* |
| *Expected Output* | *3 groups of pins of Butterfree show up at the corresponding location* |
| *Test steps* | *1. Scroll down the list until the Pokémon is found.*<br>*2. Select the Pokémon* |
| *Actual Output* | *Matches expected output* |
| *Result* | *Passed* |